SupCon-ViT: Supervised contrastive learning for ultra-fine-grained visual categorization

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***Abstract*—With the increasing availability of datasets exhibiting fine granularity and subtle differences between categories, fine-grained visual categorization tasks have gained significant attention across various domains. However, the focus often lies solely on overall dataset performance metrics such as top-1 accuracy, while lacking a comprehensive understanding of the underlying factors. This paper addresses this gap by presenting a detailed analysis of the CUB-200-2011 dataset through extensive experiments. We identify and investigate specific ultra-fine-grained subsets that significantly impact the overall accuracy of the dataset. To enhance the performance of ultra-fine-grained visual classification, we propose SupCon-ViT, an ultra-fine-grained visual categorization network based on supervised contrastive learning. The key component of our approach is a supervised contrastive learning module, which effectively guides the network to learn discriminative local features within samples. This is accomplished by continuously pulling closer the normalized embeddings from the same class and pushing away embeddings from different classes. As a result, our approach achieves discriminative local representations, leading to improved network classification performance. Experimental results demonstrate the effectiveness of our proposed method on four ultra-fine-grained subsets of the CUB dataset. Notably, our approach achieves significant performance improvements without requiring additional expert information during training. This work contributes to the broader understanding of fine-grained visual categorization and offers a practical solution to enhance the accuracy of ultra-fine-grained visual classification tasks.The code is available at https://github.com/lucinda0love/SupCon-ViT-pytorch.**

***Index Terms*—Ultra-fine-grained visual categorization, Supervised contrastive learning, Transformer**

I. Introduction

With the advent and ongoing development of deep learning, copious amounts of data have become available, leading to the emergence of numerous branches within digital image processing and learning. One such branch is fine-grained image recognition [1], [2], which aims to achieve more precise classification of subordinate categories that often exhibit large intra-class differences and small inter-class differences [3]. To accomplish this, models must not only consider the overall sample but also learn its local features in order to accurately classify these subordinate categories.

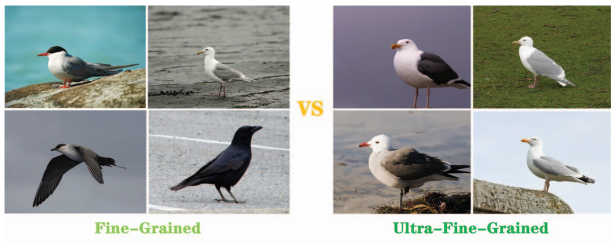

Fig. 1. An example of fine-grained samples in comparison with ultra-fine-grained samples from the CUB-200-2011 [4]. Each sample represents a category in fine-grained and every sample represents a subordinate category in ultra-fine-grained.

Although discriminating fine-grained samples remains a challenging task for the human naked eye, deep neural networks have made significant strides in fine-grained recognition. Some state-of-the-art (SOTA) models have achieved top-1 accuracy of over 90% [5]–[8] on commonly used public datasets such as CUB-200-2011 [4] (see Fig. 1), FGVC-Aircraft [9], and Stanford Cars [10]. However, no research has focused on the samples that lower the accuracy, and the dataset remains a black box to us. What are the primary misclassified samples? Upon analysis, we found that the misclassified samples are mostly assigned to closely related subordinate classes with small inter-class differences, which belong to the category of ultra-fine-grained data. Fig. 1 shows an example of visually comparing from fine-grained samples to ultra-fine-grained samples. In contrast to fine-grained samples, ultra-fine-grained samples are more difficult to be distinguished which own extremely large inter-class similarity.

In this paper, we test three representative models on the CUB [4] (see Fig. 1) to avoid models of the same type producing the same results, which has certain limitations. The three models we test are the traditional but highly-cited Resnet50 [11], MaskCOV [5], which has a self-supervised data augmentation module, and ViT-B_16 [12], which is based on the Transformer architecture rather than a convolutional neural network. By completing a series of experiments, we obtain the results shown in Table I, which indicates that the accuracy for this type of data is significantly lower than the average.

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TABLE I
HERE ARE THE CLASSIFICATION ACCURACIES OF DIFFERENT MODELS ON CUB [4] AND ITS SUBSET CUB_CROW. CUB/CROW REPRESENTS THE ACCURACY ACHIEVED BY USING THE ENTIRE CUB AS THE TRAINING SET, CLASSIFICATION ACCURACY ON THE TESTING SET OF THE ENTIRE CUB AND ON THE TESTING SET OF THE SUBSET CUB_CROW.CUB_CROW REPRESENTS THE ACCURACY ACHIEVED BY USING ONLY THE SUBSET CUB_CROW AS THE TRAINING SET.

Method	Backbone	Top-1 Accuracy (%)	
		CUB/Crow	CUB_Crow
Resnet-50	Resnet-50	85.5/59.1	63.2
MaskCOV	Resnet-50	86.6/60.1	62.1
ViT-B_16	Transformer	90.6/77.7	81.8

This suggests that models which are suitable for fine-grained datasets may not necessarily be suitable for ultra-fine-grained datasets.

We conducted a series of experiments and found that training a model only on the ultra-fine-grained subset from a fine-grained dataset can improve the accuracy of recognizing similar samples. However, the improvement is not significant enough to reach the average level of accuracy, as demonstrated in Table I. We used a subset of the CUB [4], named CUB_Crow as an example. Interestingly, when the dataset becomes smaller, the number of model parameters increases, and the classification accuracy may improve, but this improvement is insufficient, indicating the difficulty of recognizing ultra-fine-grained subsets. The recognition method used for fine-grained datasets may not be effective enough for such subsets, calling for the development of specialized techniques and models.

The reason for the impact on the classification accuracy of fine-grained datasets is due to the presence of ultra-fine-grained subordinate categories. In this paper, we conducted a detailed analysis of the CUB [4] and selected ten ultra-fine-grained subsets for further study (as shown in Table II). The question now becomes why these ultra-fine-grained subsets affect the overall accuracy and why they are difficult to distinguish from one another, so what unique properties they possess? Compared to fine-grained samples, ultra-fine-grained samples have a higher intra-class variance and a smaller inter-class variance [5], [6], [13] (see Fig. 2), making them almost indistinguishable to the naked eye and requiring the capture of subtle local features for differentiation. Moreover, each highly fine-grained subcategory often only contains a few samples [6], which further increases the difficulty of classification.

Existing methods to address this problem involve adding data augmentation modules [5], [6] to increase sample diversity and facilitate the exploration of sample features, which can capture subtle features. However, designing effective data augmentation strategies can be inconvenient and may carry some risks. In most cases, the network may not be able to explore local features independently and instead focuses on the overall information of the samples. To address this issue, we propose SupCon-ViT, a supervised contrastive learning-

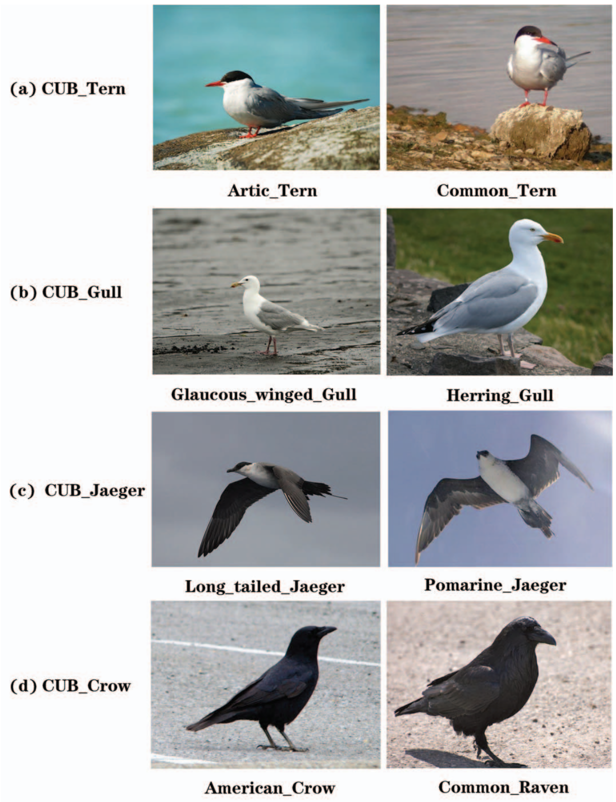

Fig. 2. Here is an example of four subsets of the CUB for ultra-fine-grained classification, where each row represents a subset, and each column represents a sample from a subordinate category within that subset.

based network for ultra-fine-grained visual categorization. By introducing a supervised contrastive learning (SupCon) module that uses encoded representations, we can widen the inter-class distance and narrow the intra-class variance, guiding the network to learn the local features of the samples. Specifically, SupCon-ViT uses a simple supervised contrastive learning (SupCon) module that provides additional supervision signals to differentiate sample categories in conjunction with label supervision. We conduct experiments on the ultra-fine-grained subsets of CUB, and the results show that this approach can improve the network's performance on ultra-fine-grained visual classification tasks and enhance its robustness. In summary, the main contributions of this paper are as follows:

1) : We conduct a detailed analysis of CUB [4] and select ten ultra-fine-grained subsets that affect the overall top-1 accuracy on CUB [4].

2) : We propose a supervised contrastive learning-based network (SupCon-ViT) for ultra-fine-grained visual categorization, which effectively widens the inter-class variance and narrows the intra-class variance and focus more on those integral and informative regions.

3) : We conduct experiments and comprehensive analysis using network we proposed on the four most challenging ultra-

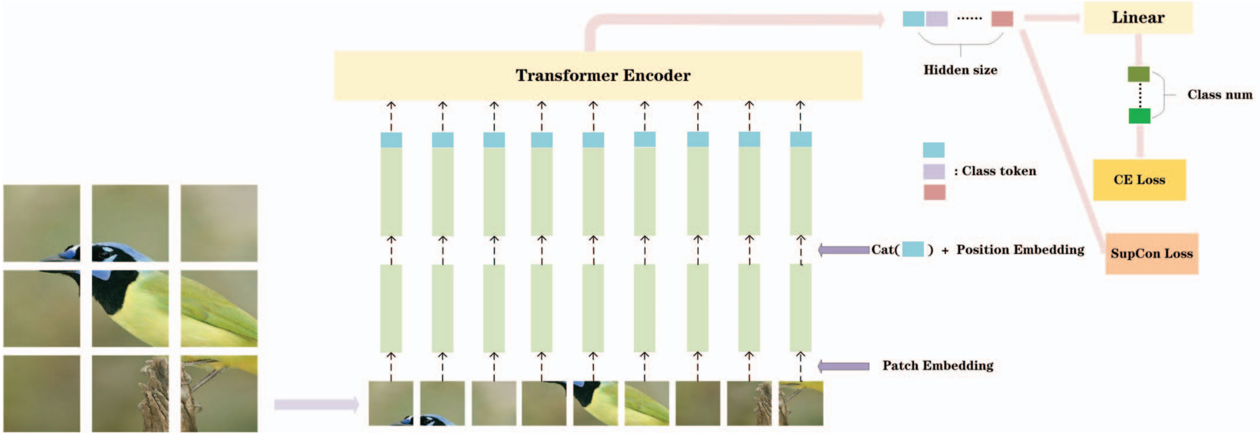

Fig. 3. An overview of the proposed SupCon-ViT. An input image is splitted according to the size of patch, linearly embed each of them, concatenate classification tokens, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer Encoder. Finally, the obtained classification tokens are used for CE loss and SupCon loss.

fine-grained subsets in the CUB.

II. Related Work

A. Ultra-fine-grained visual categorization

The continuous development of deep learning [14] has brought about a revolutionary transformation in computer vision-related tasks. Visual classification tasks are relatively mature, and fine-grained visual classification tasks have been applied in various fields, such as remote sensing image analysis [15], [16], biomedical information processing [17], [18], and face analysis and modeling [19]–[21]. Ultra-fine-grained visual classification is a subtask of fine-grained visual classification, with greater difficulty, fewer samples, and smaller inter-class variance but larger intra-class variance [13], [22]–[24], making it challenging for even human experts to quickly and accurately identify. It is undoubtedly a highly challenging task.

To address the forementioned challenges, [5] proposed a MaskCOV network that uses a random mask covariance module to learn discriminative representations for ultra-fine-grained visual classification. They added a data augmentation module as an in-image data augmentation and guide the model to focus on local regions with discriminative features. [25] proposed a self-supervised part erasing framework called SPARE, which encourages the network to explore the intrinsic structure of the data and locate discriminative representations, enabling it to learn both the global and local features of the data. [6] proposed a mixed attentive vision transformer called Mix-ViT, which combines tokens from different semantic levels to enable the network to learn and predict whether tokens are replaced, forcing it to understand the highly discriminative details of the data.

B. Contrastive learning

The main idea of contrastive learning [26] is to construct positive and negative pairs through a pretext task for model training. Ideally, the distance between positive and negative samples should be large, and the distance between positive samples should be relatively small. With this idea, a loss function can be designed to enable the model to learn the mapping relationship between data in a high-dimensional vector space. Contrastive learning [26] has shown great potential in unsupervised clustering algorithms [27], [28], and has been successfully used in natural language processing models such as GPT [29] and BERT [30]. It has also been applied to image classification tasks, such as SimCLR [31] and MoCo [32], which use self-supervised contrastive learning to achieve classification tasks. Supervised contrastive learning [33] is an extension of contrastive learning that uses data label information to distinguish between positive and negative samples, making it more effective in utilizing label information while gathering feature points of the same class together and dispersing feature points of different classes.

Supervised contrastive learning (SupCon) enables label information to be reused and incorporated into the model training process, making it more effective in utilizing the available information [34]–[36]. By using label information to distinguish positive and negative sample pairs, the model can learn more discriminative features and better capture the intrinsic structure of the data. This can lead to improved performance on various classification tasks, including ultra-fine-grained visual categorization.

III. Method

As we mentioned before, inter-class variances from ultra-fine-grained objects are extremely subtle, thus requiring highly discriminative representation for effective classification. In order to address this problem, it's necessary to gain more knowledge from the intrinsic structure of input data while effectively distinguish ultra-fine-grained objects from the same class or not. Therefore, we are motivated to design a supervised contrastive learning module (see Fig. 3), which can help network pulling closer normalized embeddings from the same class and push away the embeddings from different classes.

A. Feature learning

Following ViT [12], We begin with the input data $I \in \mathbf{R}^{H\times W}$ (channels is not considered), where H, W denote height and width of original image, then split it to fixed-size $N \times N$ patches denoted by I_i, where i is indice and $1 \leq i \leq M$, and $M = HW/N^2$.

After that, M patches are feed to Vision Transformer, we first flatten every patch and map it to D dimensions $\mathbf{x}_i \in \mathbf{R}^{1\times D}$ with a convolution kernel called trainable linear projection, where D is a scalar named hidden size we define and can be adjusted, so we get matrix $\mathbf{X}$ of size $M \times D$. Then an extra classfication token $\mathbf{c} \in \mathbf{R}^{1\times D}$ is concatenated to as classfication information and undertakes classfication task, followed by a learnable vector called position embedding is added. Therefore, the whole process is called Transformer Embedding, the overall embedding operation is defined as follows:

$$\mathbf{z}_e = \begin{pmatrix} \mathbf{c} \\ \mathbf{x}_i \\ \vdots \\ \mathbf{x}_M \end{pmatrix} + \mathbf{E}_{pos}, \tag{1}$$

where $\mathbf{E}_{pos} \in \mathbf{R}^{(M+1)\times D}$ is position embedding matrix.

The resulting sequence of embedding vectors $\mathbf{z}_e$ serves as input to the Transformer Encoder which consists of a certain amount of multi-head self-attention (MSA) and multi-layer perceptron (MLP) blocks, which can be formatted as follows:

$$\mathbf{z}_{l\_attention} = MSA\left(LN\left(\mathbf{z}_{l-1}\right)\right) + \mathbf{z}_{l-1}, \tag{2}$$

$$\mathbf{z}_l = MLP\left(LN\left(\mathbf{z}_{l\_attention}\right)\right) + \mathbf{z}_{l\_attention}, \tag{3}$$

where $LN(\cdot)$ is the layer normalization operation and $\mathbf{z}_{l\_attention}$ denotes the encoded image representation after attention operation. Similarly, $\mathbf{z}_l$ denotes the final encoded image representation in l-th encoder layer. After L blocks, we can obtain the classfication token vector.

B. Supervised contrastive learning

This section presents the proposed supervised contrastive learning module. Specifically, the classification token vector in every batch can be denoted as:

$$\mathbf{c}_j = \begin{bmatrix} c_{j1} & \dots & c_{jD} \end{bmatrix}, \tag{4}$$

where $\mathbf{c}_j$ denotes class token of j-th image in a batch, and D is hidden size.

Those classification tokens is further propagated through Transformer Encoder which like a projection network, so they are representative of input images. Therefore, we can use them to calculate the dot product between encoded representation vector of samples in every batch, which can be thought of similarity. Via calculating the dot product between every pair of samples, meanwhile, combined with labels of samples, the supervised contrastive losses can be obtained. Particularly, those with the same label are considered positives, otherwise they are negatives. Supervised contrastive losses in every batch can takes the following form:

$$\begin{aligned} L_{SupCon}^{batch} &= \sum_{b\in I_{batch}} L_{SupCon,b}^{batch} = \\ &\sum_{b\in I_{batch}} \frac{-1}{|P(b)|} \sum_{p\in P(b)} \log \frac{exp\left(\mathbf{c}_b \cdot \mathbf{c}_p/\tau\right)}{\sum\limits_{a\in A(b)} exp\left(\mathbf{c}_b \cdot \mathbf{c}_a/\tau\right)}, \end{aligned} \tag{5}$$

where $P(b)$ is the set of postives, $A(b)$ is the set of all samples in a batch, I_{batch} denotes the set of all samples except for b in a batch, and τ is a scalar temperature parameter. After calculating supervised contrastive losses of all batches, the sum is averaged to get the final supervised contrastive loss as follows:

$$L_{SupCon} = \frac{1}{n\_bs} \sum_{q=1}^{n\_bs} L_{SupCon}^{q}, \tag{6}$$

where $n\_bs = ceil(n/batch\_size)$ denotes the number of batches, n is the number of samples in training set, and $ceil(\cdot)$ is ceil function.

C. Classification loss

Like supervised contrastive learning section, the classification token from Eq. 4 is also used as the classification information, then a classification head is applied to the classification token to predict whether each sample is classified correctly.

$$\mathbf{p}\left(\mathbf{c}\right) = softmax\left(\mathbf{c} \cdot \mathbf{W}_{class}\right), \tag{7}$$

where $\mathbf{c}$ is the classification token and $\mathbf{W}_{class} \in \mathbf{R}^{D\times K}$ is the classification matrix, K denotes the number of classes. We then can define a classification loss function L_{cls} as:

$$L_{cls} = -\widetilde{\mathbf{y}} \cdot \log\left(\mathbf{p}\left(\mathbf{c}\right)\right), \tag{8}$$

where $\widetilde{\mathbf{y}}$ is an one-hot vector indicating the associated groundtruth category label. So the total loss in the proposed framework is defined as:

$$L = \alpha L_{cls} + \beta L_{SupCon}, \tag{9}$$

where α and β are the hyper-parameters weighting the contributions from L_{cls} and L_{SupCon}, respectively. The selection of α and β is discussed in the experiments (see Table V).

IV. Experiments

In this section, we mainly introduce the datasets we used and implementation details. Next, performance evaluation are discussed on ultra-fine-grained datasets we selected.

A. Datasets

The proposed method was subjected to a series of experiments on the CUB-200-2011 (CUB), which consists of 200 categories with approximately 60 samples per category. The dataset is split into training set and testing set with a ratio of 1:1. We also conduct an analysis of the dataset and select 10 ultra-fine-grained subsets, as shown in Table II. Finally,

TABLE II
THE TOP-1 ACCURACY(%) OF THE ULTRA-FINE-GRAINED SUBSET IN THE CUB. THE NUMBER IN THE PARENTHESES OF EACH SUBSET REPRESENTS THE TOTAL NUMBER OF CATEGORIES OF THE SUBSET. THE SUBSET HERE IS NAMED ACCORDING TO ITS PARENT CATEGORY. FOR EXAMPLE, CUB_JAEGER (2) INCLUDES 071.LONG_TAILED_JAEGER, 072.POMARINE_JAEGER THESE TWO SUBORDINATE CATEGORIES. THE ITALICS ARE THE MOST DIFFICULT SUBSETS WHICH ARE USED FOR FOLLOW-UP EXPERIMENTS.

CUB	*CUB_tern (7)*	*CUB_gull (8)*	CUB_sparrow (21)	CUB_swallow (4)	CUB_cormorant (3)
86.6	*64.1*	*61.1*	76.9	75.8	75.3
CUB	*CUB_jaeger (2)*	*CUB_crow (9)*	CUB_flycatcher (7)	CUB_shrike (2)	CUB_hummingbird (3)
86.6	*56.7*	*60.1*	74.4	70.0	78.9

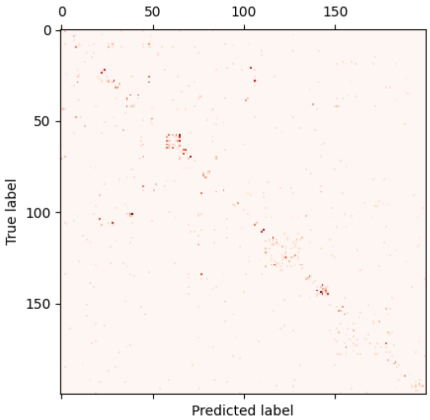

Fig. 4. Confusion matrix of misclassified samples of CUB test set on MaskCOV

experiments are conducted on the most challenging subset (indicated in italics in Table II), where the top-1 accuracy of four subsets is below 70%. The specific method of subset selection is as follows:

Based on the results presented in Table I, we use the MaskCOV, which has a middling top-1 accuracy on the entire CUB but the lowest accuracy on the CUB_Crow subset, as the baseline model for analyzing the CUB. MaskCOV [5] achieves a top-1 accuracy of 86.6% on the entire CUB. By examining the confusion matrix (see Fig. 4) of misclassified samples in the testing set, we observe that a large number of misclassifications occur near the diagonal of the confusion matrix. In the CUB [4], subcategories belonging to the same category are numbered consecutively, meaning that most of the misclassified samples are classified into the same category. We extract the subcategories with a top-1 accuracy lower than 80% and identify the categories that affect the overall classification accuracy, which are the ultra-fine-grained subcategories and are shown in Table II.

The reason why CUB_Crow subset has 9 categories is that there are some highly similar samples to the 4 Crow categories (see Table III column Crow), which belong to the 5 categories (see Table III column Others). Considering that there are a considerable number of these samples, they were grouped together with the crow category to form the CUB_Crow subset, which has a total of 9 categories. The other 9 subsets are completely based on the same species but different subordinate categories, so there is no situation similar to the CUB_Crow subset.

TABLE III
DETAILS ON CUB_CROW.

Crow	Others
029.American_Crow	004.Groove_billed_Ani
030.Fish_Crow	010.Red_winged_Blackbird
107.Common_Raven	023.Brandt_Cormorant
108.White_necked_Raven	027.Shiny_Cowbird
	049.Boat_tailed_Grackle

B. Implementation Details

1) Experiment Environment: The network we proposed is implemented on PyTorch. The computer environment for the experiment is as follows: Python version is 3.10.6, PyTorch version is 1.12.1, the graphics card is 'NVIDIA GeForce RTX 3060' with 'CUDA version 11.6'.

2) Backbone network: Unlike convolutional neural networks, Vision Transformers [12] have shown outstanding performance, so we use it as our baseline method. Since the version ViT-B_16 with the fewest parameters has already demonstrated strong performance and our resources are limited, we directly use it without using other versions with more parameters. ViT-B_16 has 12 layers, 12 heads, a hidden size of 768, an MLP size of 3072, an input patch size of 16×16, and it has been pre-trained on the ImageNet [37]. The input images are resized to a fixed size of 440 × 440 and randomly cropped into 384 × 384. Data augmentation is applied to each input image in the training set using the first 8 of the 25 sub-strategies provided by AutoAugment [38], while the test set is not augmented. These settings remain constant throughout the entire experiment.

3) Supervised contrastive learning (SupCon) module: To keep the model simple, we did not add any additional parameter layers and directly used the classification token from the last layer of backbone network, which is an encoded representation, to calculate the supervised contrastive learning loss.

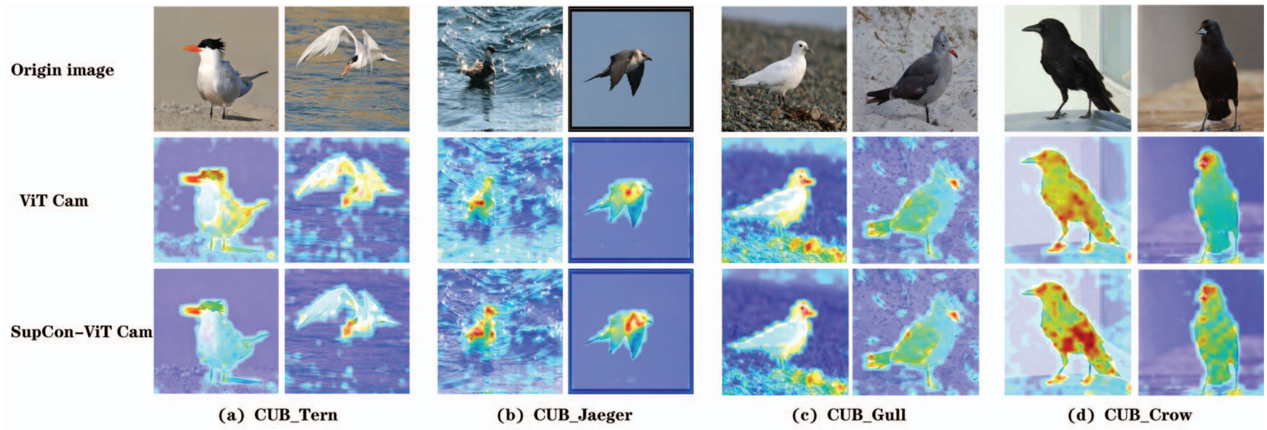

Fig. 5. An example of illustrating attention maps obtained by ViT and SupCon-ViT where highlighted regions are discriminative for classification.

TABLE IV
THE TOP-1 ACCURACY OF THE ULTRA-FINE-GRAINED SUBSETS IN CUB AND CUB UNDER DIFFERENT MODELS, VIT (CUB) MEANS USING CUB FOR MODEL TRAINING AND THOSE ULTRA-FINE-GRAINED SUBSETS FOR MODEL TESTING. SIMILARLY, VIT AND SUPCON-VIT MEANS USING ULTRA-FINE-GRAINED SUBSETS FOR BOTH MODEL TRAINING AND TESTING.

Method	Top-1 Accuracy (%)			
	CUB_Crow	CUB_Gull	CUB_Tern	CUB_Jaeger
ViT(CUB)	77.7	71.2	72.7	75.0
ViT	81.8	71.2	67.9	76.7
SupCon-ViT	**82.9**	**71.6**	**71.3**	**78.3**

4) Training and inference: All experiments in this paper with ViT as the backbone network are trained for 20000 iterations using SGD with a batch size of 10, and the learning rate is 0.003 initially.

C. Evaluation on ultra-fine-grained subsets of CUB

To validate the capability of the proposed SupCon-ViT in the ultra-fine-grained visual categorization tasks, we first conduct evaluations on the four ultra-fine-grained subsets. Table IV lists the top-1 classification accuracy of the competing method and proposed network SupCon-ViT. The proposed SupCon-ViT achieves higher classification accuracy than its backbone network ViT on four ultra-fine-grained subsets CUB_Crow 1.1%, CUB_Gull 0.4%, CUB_Tern 3.4%, CUB_Jaeger 1.6%, respectively, so it outperforms this very popular method recently (ViT) with a considerable margin. That sufficiently demonstrates its effectiveness on ultra-fine-grained visual categorization tasks. This result also verify the proposed SupCon-ViT has potential towards the challenging ultra-fine-grained visual categorization tasks.

Meanwhile, Table IV also lists the differences between a large dataset (CUB) and its small subset (four ultra-fine-grained subsets) on the same model (ViT). We can observe that classification accuracy using subset for training is not lower than those using whole dataset for training on three out of four subsets (CUB_Crow, CUB_Gull, CUB_Jaeger). That's verifies what we mentioned in Section I. When the dataset becomes smaller, the number of model parameters increases relatively, and the classification accuracy may improve.

D. Ablation study

1) Visualization: To further demonstrate the effectiveness of SupCon-ViT, we visualize the attention maps of ultra-fine-grained subset in Fig. 5, where highlighted regions contribute significantly to the visual categorization. We noticed that the attentive regions of different subsets are diverse. The highlighted regions of CUB_Crow and CUB_Jaeger are distributed along the texture and color of the feathers. On the CUB_Tern and CUB_Gull subsets, the attentive regions accurately cover beaks.

Surprisingly, we observe that birds' whole body is covered in all attention maps, besides, those representational parts are highlighted. This is consistent with human observations that first the whole and then the local features, which is vital for the determination of bird species. The results also demonstrate the potential of the proposed method towards ultra-fine-grained visual categorization task.

We also compare the differences of attention maps on between ViT and the proposed SupCon-ViT. We notice there exists some subtle differences. SupCon-ViT can pay more attention to those discriminative regions, where is highlighted more darker than attention maps on ViT. Meanwhile, SupCon-ViT can do better to ignore those parts which is less meaningful for distinguish targets. This can further prove the performance improvement of the proposed SupCon-ViT.

2) Ablation on Weight of Loss Terms: Table V lists the performance with ratios of weight parameters α and β. We only use the CUB_Crow subset as an example. The classification accuracy of the proposed method SupCon-ViT under CUB_Crow increase significantly as β increases and α keeps unchanged. This result remains same as the ratio keeps reducing to 1:1.5, but it decreases if the ratio continues

TABLE V
CLASSIFICATION ACCURACY WITH DIFFERENT WEIGHT RATIOS OF LOSS TERMS. THE FINAL CHOICE IS IN BOLDFACE.

$\alpha : \beta$	Top-1 Accuracy (%)
	CUB_Crow
2:1	81.9
1:1	82.9
1:1.3	**82.9**
1:1.5	82.9
1:2	81.4

reducing to 1:2. So we select the middle one 1:1.3 for later experiments. Despite more diversified combinations of α and β may further improve the performance, we roughly set 5 combinations for simplicity and verify such a rough setting is enough to demonstrate the effectiveness of the proposed SupCon-ViT.

V. CONCLUSION

This paper conducts a detailed analysis of CUB, uses its classification accuracy as the criterion, and finds out the ultra-fine-grained subsets that lower the overall accuracy through experiments.We develop a framework, SupCon-ViT, for ultra-fine-grained visual categorization task. Supervised constrative learning (SupCon) module is used to drive the model to distinguish the details among inter-class samples. At the same time, SupCon module can effectively push away the samples from different classes and pull closer those samples from the same classes. SupCon-ViT has achieved comparable performance on four ultra-fine-grained subsets of CUB. This also verifies supervised constrative learning (SupCon) is a promising module for challenging ultra-fine-grained visual categorization.

VI. ACKNOWLEDGEMENT

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